

NovaPharma Iberia — Step 4.A Review: Power BI Model

Digital transformation project · Excel → MySQL / Python / Power BI · Author: Nix · 25 September 2026

4.0 What we show, and why

Before connecting anything, we decided what the report should answer. The Excel workbook monitored a lot: 17 sheets, a dashboard, consolidation grids and budget comparisons. We keep only the essential, the same questions a manager asks every month:

- **How are we doing this year?** Net sales YTD, gross margin %, market share, and a comparison with budget.
- **How is it trending?** Net sales month by month over the last 12 months.
- **Where does it come from?** By brand (with drill-down to SKU) and by region.

That becomes two pages plus an About page:

Page	Content
1 Overview	4 KPI cards, a 12-month trend line, net sales by brand and a region table
2 Budget vs Actual	Kept deliberately simple (1–2 visuals), just enough to cover what the Excel did
About	What the report is, where the data comes from and when it was last refreshed

Everything else from the Excel stays in a backlog: channel × therapeutic area matrix, secondary cards, top-5 tables.

4.A The model

Two SQL views as the bridge. Instead of rebuilding the logic in Power BI, we added two views in MySQL (03_SQL/02_views.sql):

- **v_sales_enriched:** every sale with its product, brand, customer, territory and month, plus net sales, COGS and gross margin, calculated once in SQL.
- **v_bva:** actuals and plan in one long table, so they can be compared side by side.

The rule behind this: **SQL calculates what is fixed per row, Power BI calculates what depends on the user's selection.**

Import. Power BI connects to novapharma_dw (MySQL, port 3307) in Import mode. 9 objects are loaded: the 2 views, fact_market and 6 dimensions.

Relationships. Five came automatically from the foreign keys in MySQL. The views have no foreign keys, so the other six were drawn by hand. All of them are many-to-one, with the filter going from the dimension to the fact:

Fact	→ Dimension (key)
v_sales_enriched	dim_calendar (month_key), dim_product (sku_code), dim_customer (customer_code)
v_bva	dim_calendar (month_key), dim_product (sku_code), dim_territory (territory_code)

Two links were left out on purpose:

- **Sales → brand / territory.** The model is a snowflake: sales already reach brand through product and territory through customer. A direct link would create a second path between the same tables, and Power BI allows only one active path. Lesson: check the paths, not just the matching columns.
- **v_bva → dim_version.** The 'Actual' version doesn't exist in dim_version, so it would show up as (Blank). The version filter uses the column inside v_bva instead.

Validation. Total net sales in Power BI = **442.94M**, the same as in SQL (442,941,738.80). A slicer on region (ES-North → 78.88M) confirmed that the filter travels territory → customer → sales.

Result: 11 relationships, one path between any two tables, and totals that match the database. The model is ready for the measures (4.B) and the visuals (4.C).

Vocabulary

Snowflake: a model where a dimension reaches the fact through another dimension (brand → product → sales).

Slicer (Segmentación de datos): a visual that filters the whole page. Filter context: the set of filters active on a number (slicers, axes, rows). Backlog: the list of things left out on purpose, to do later if needed.